Assignment 3

1	s = "world" for char in reversed(s): print(char, end='')

2	sum = 0
	i = 0
	while i <= 100:
	  if i % 2 == 0:
	  sum += i
	  i += 1
	print(sum)

3	for i in range(1, 31):
	  if i % 3 == 0:
	  print(i)

4	for i in range(5):
	  for j in range(5):
	  print('*', end=' ')
	  print()

5	d = {} for i in range(97, 102): d[chr(i)] = i print(d)
6	s = "a,b,c,d,e" items = s.split(',') for item in items: print(item.upper())
7	n = 0 while n**2 <= 100: n += 1 print(n)

8	d = {1: 'one', 2: 'two', 3: 'three'}
	for key in d:
	  if key % 2 == 0:
	  print(d[key])

9	for i in range(1, 51):
	  if i % 5 == 0 and i % 7 == 0:
	  print("FizzBuzz")
	  elif i % 5 == 0:
	  print("Fizz")
	  elif i % 7 == 0:
	  print("Buzz")

10	numbers = []
	for i in range(1, 11):
	  numbers.append(i**2)
	for number in numbers:
	  print(number)

11	output = ''.join([chr(num) for num in range(100, 110) if num % 3 == 0]) print(output)

12	data = {"apple": 10, "banana": 5, "cherry": 15}
	while data:
	  key, value = data.popitem()
	  if value % 2 != 0:

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        print(f"{key}: {value}")
```

13
```
i = 10
while i > 0:
    i -= 3
    print(i, end=' ')
    if i % 4 == 0:
        break
```

14
```
matrix = [[x for x in range(y, y+3)] for y in range(0, 9, 3)]
print(matrix[1][1] + matrix[2][0])
```

15
```
def modify_list(lst):
    for i in range(len(lst)):
        if lst[i] % 2 == 0:
            lst[i] //= 2
        else:
            lst[i] *= 2
numbers = [1, 2, 3, 4, 5]
modify_list(numbers)
print(numbers)
```

16
```
fibonacci = [0, 1]
for i in range(2, 10):
    next_value = fibonacci[-1] + fibonacci[-2]
    if next_value % 5 == 0:
        continue
    fibonacci.append(next_value)
print(fibonacci)
```

17
```
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True
primes = [x for x in range(50) if is_prime(x)]
print(len(primes))
```

18
```
def check_sequence(lst):
    return all(x < y for x, y in zip(lst, lst[1:]))
num_list = [5, 10, 15, 20, 25]
print(check_sequence(num_list))
```

19
```
class Dog:
    def __init__(self, name):
        self.name = name
    def speak(self):
        return f"{self.name} says Woof!"
buddy = Dog("Buddy")
print(buddy.speak())
```

20
```
s = {i for i in range(1, 11) if i % 2 == 0}
filtered_set = {x for x in s if x > 5}
print(filtered_set)
```